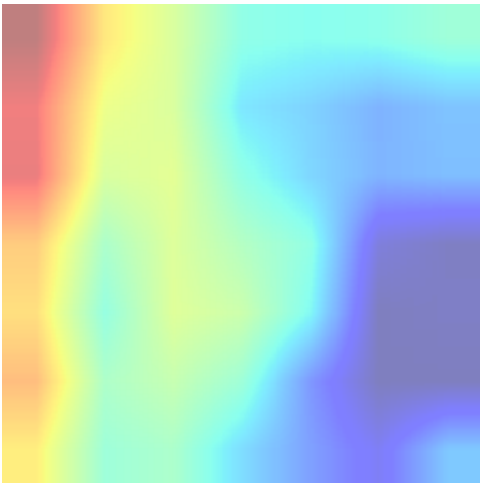


Clinical Diagnostic & RAG Report — Breast Cancer

Target Domain:	Breast Cancer	Image File:	PIL_Image
Predicted Class:	malignant	Confidence:	66.0%
95% Conformal Set:	['malignant']	Uncertainty Status:	■ High Confidence Singleton

Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan

Grad-CAM Saliency Overlay (ROI)

AI Clinical Explanation (FLAN-T5 Grounded RAG)

"The image was classified as 'malignant' with 66.0% confidence. Biomedical literature (PMID:35391913) indicates: The proposed method has achieved state-of-the-art results, with an accuracy of 0.9070 and the area under the receiver operating characteristic curve of 0.9459. The results demonstrate that the proposed method has the potential to be used in the clinical diagnosis of benign and malignant breast cancer."

Faithfulness & Quantitative Text Metrics

NLI Faithfulness Score:	0.3338 / 1.0000	ROUGE-1:	0.8000
BLEU-4:	0.7636	BERTScore Similarity:	0.5631

Retrieved PubMed Literature Evidence

[1] Ultrasound Image Texture Feature Learning-Based Breast Cancer Benign and Malignant Classification. (PMID: 35391913)

"The proposed method has achieved state-of-the-art results, with an accuracy of 0.9070 and the area under the receiver operating characteristic curve of 0.9459. The results demonstrate that the proposed method has the potential to be used in the clinical diagnosis of benign and malignant breast cancer."

[2] Diagnosis of Benign and Malignant Breast Nodules by Conventional Ultrasound in Combination with S-Detect Technology and Elastic Imaging. (PMID: 39410680)

"CONCLUSION: The combination of conventional ultrasound BI-RADS classification with elasticity score or S-Detect technology has a higher diagnostic efficacy for breast nodules, which can improve breast cancer detection and provide valuable diagnostic evidence for clinical practice. KEY WORDS: S-Detect technology, Ultrasound elastic imaging, Elasticity scoring, Elasticity strain ratio value, Breast tumour."

[3] Ultrasensitive Ultrasound Microvessel Imaging for Characterizing Benign and Malignant Breast Tumors. (PMID: 31530420)

"To evaluate the performance of UMI, we studied 44 patients with 51 breast masses. Tumor pathology served as the gold standard: 28 malignancies and 23 benignities. UMI provided a significant improvement in depicting smaller vessels compared with conventional Doppler."

[4] Breast Tumor Classification Using Intratumoral Quantitative Ultrasound Descriptors. (PMID: 35295204)

"The results indicate that the identified features are able to distinguish between benign and malignant breast lesions very effectively, and the combination of the features identified in this research has the potential to be a significant tool in the noninvasive rapid and accurate diagnosis of breast cancer."

[5] Classification of malignant tumours in breast ultrasound using unsupervised machine learning approaches. (PMID: 33446841)

"Here, a new workflow, utilising machine learning to recognise malignant US images was proposed. Comparison of physician diagnoses and the automatic classifications made using machine learning yielded similar outcomes. This indicates the potential applicability of machine learning in clinical diagnoses."